

#> RNN Recurrent Neural Networks

ANN $\rightarrow$ use in tabular data.

CNN $\rightarrow$ images (Grid data).

RNN $\rightarrow$ is used in sequential data. (NLP).

$\Rightarrow$ RNN is type of sequential model to work on sequential data.

Ex:-

iq marks gender

$\Rightarrow$ if you switch in place of iq to marks to gender. It don't effect. becz it is not sequential. but when we talk about text.

Hi my name is Rajan. if you switch words each other grammatically it is incorrect.

$\Rightarrow$ Time series data, Speech, DNA Sequence.

#> Why use RNN?

Problem using ANN in sequential data.

1. text input $\rightarrow$ Varying Size.
2. Zero padding $\rightarrow$ Unnecessary computation.
3. Prediction Problem
4. Totally disregarding the sequence info.

that why we use RNN it is perfect for Sequence data.

#> forward Propagation in RNN

#> Data for RNN :-

Data will be in (time step, input-features) form

Ex:-

Review	Sentiment
movie was good	1
movie was bad	0
movie was not good	0

Vocab 5 words

#> we can represent in vector :-

→ movie :- [1 0 0 0 0]

→ was :- [0 1 0 0 0]

→ good :- [0 0 1 0 0]

→ bad :- [0 0 0 1 0]

→ not :- [0 0 0 0 1]

review 1 [1 0 0 1] 3 words

(3, 5)
total(5)
(items)

review 2

(3, 5) total vocab (4, 5)
(items)

review 3

#> How RNN works?

Same data as we see in previous.

data :-

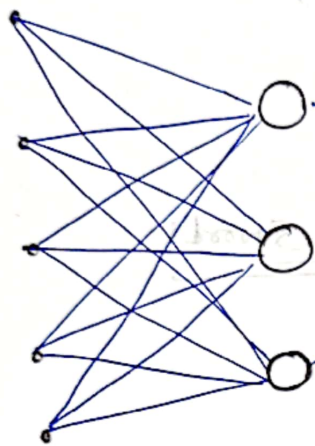
	Review	Sentiment
x_1	movie was good x_{11} x_{12} x_{13}	1
x_2	movie was bad x_{21} x_{22} x_{23}	0
x_3	movie was not good x_{31} x_{32} x_{33} x_{34}	0

Architecture of RNN

P.T.O

#> RNN Architecture :-

it's not actual picture.

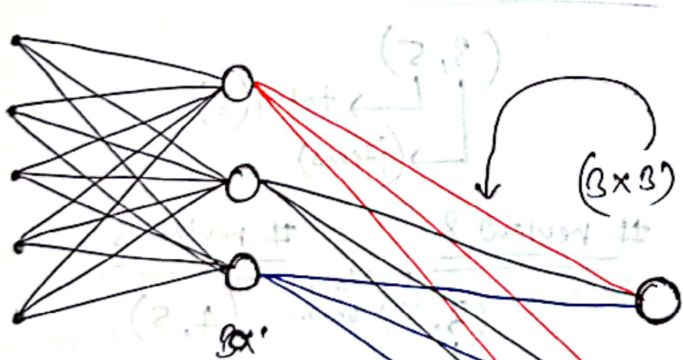


Output node (sigmoid).

at $t=1$

movie [1 0 0 0 0]

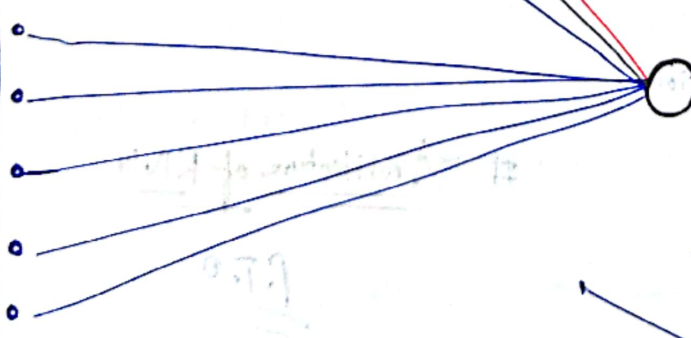
1
0
0
0
0
0



(3x3) weights

at $t=2$

0
1
0
0
0
0

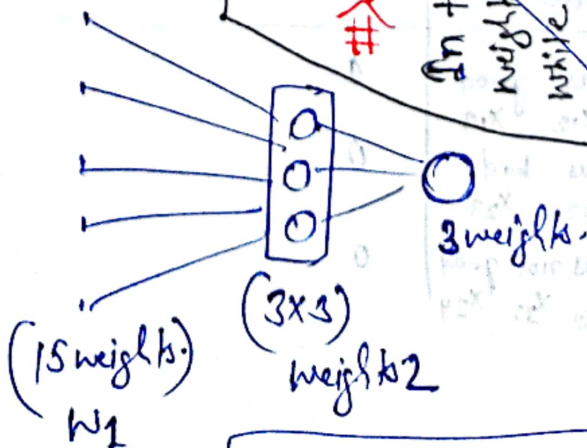


pos.

#> Summary of RNN forward prop. :-

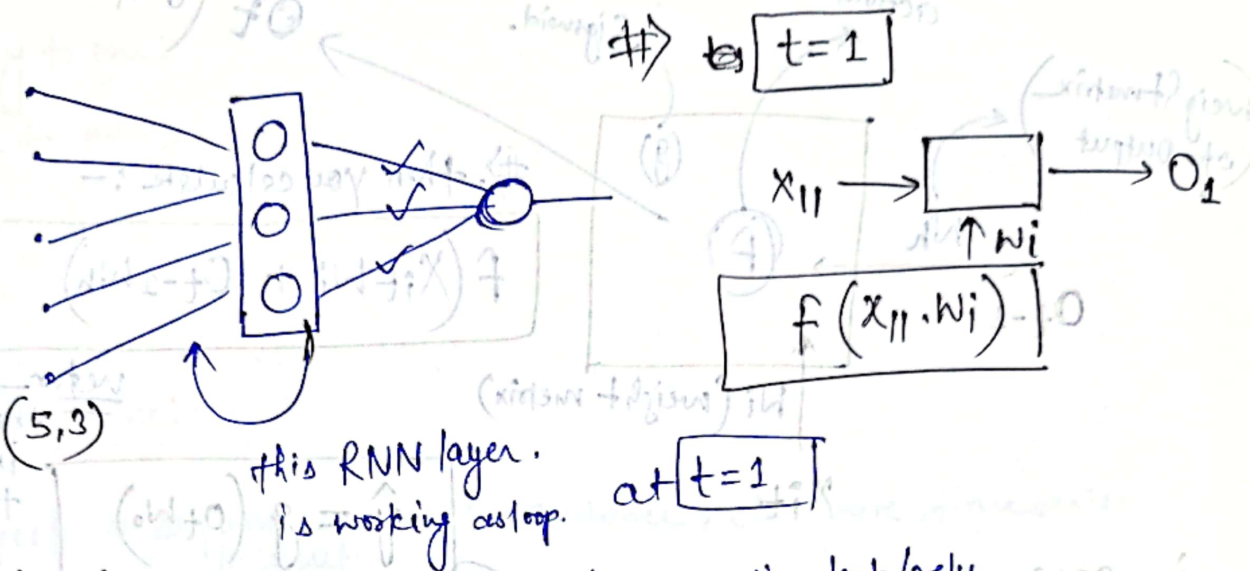
In this RNN forward prop we are using simple weights & initial weights (w_i) and get output at $t=1$ while at $t=2$ we are doing weights & initial weights or output in form of input back to hidden layer. (3x3) then we do dot product and get h_2 and so on. then we use in last activation and get final output.

incomplete



27 weights + 4 bias

RNN forward prop :-



at $t=1$ $(X_{11} \cdot W_i) \rightarrow (4,3) \rightarrow f(X_{11}, W_i)$ using activation $\tanh/\text{relu}$
 $O_1 \rightarrow \text{output } (1,3)$

at $t=2$ X_{12} O_1 comes as input in form of (W_h) with (3×3)

Diagram illustrating the RNN forward pass at $t=2$. Input X_{12} is fed into a cell along with the previous output O_1 (which acts as W_h). The cell output is O_2 .

$$X_{12} \xrightarrow{W_h} \boxed{} \xrightarrow{W_i} O_2$$

$$X_{12} \quad W_i \quad O_1 \quad W_h$$

$$(1,5) \quad (5,3) \quad (1,3) \quad (3 \times 3)$$

$$\uparrow \quad \uparrow \quad \uparrow$$

$$(1,3) \quad + \quad (1,3)$$

$$= (1,3)$$

$$f(X_{12} \cdot W_i + O_1 \cdot W_h)$$

at $t=3$ X_{13}

Diagram illustrating the RNN forward pass at $t=3$. Input X_{13} is fed into a cell along with the previous output O_2 (which acts as W_h). The cell output is O_3 .

$$X_{13} \xrightarrow{W_h} \boxed{} \xrightarrow{W_i} O_3$$

Diagram illustrating the RNN forward pass at $t=3$. Input X_{13} is fed into a cell along with the previous output O_2 (which acts as W_h). The cell output is O_3 .

$$X_{13} \quad W_i \quad O_2 \quad W_h$$

$$(1,5) \quad (5 \times 3) \quad (1,3) \quad (3 \times 3)$$

$$= (1,3)$$

then we do dot product of O_2 (W_h) weights.

Diagram illustrating the dot product calculation for O_2 and W_h .

$$(1,3) \cdot (3,1)$$

$$\uparrow \quad \uparrow$$

$$(1,1) = (1)$$

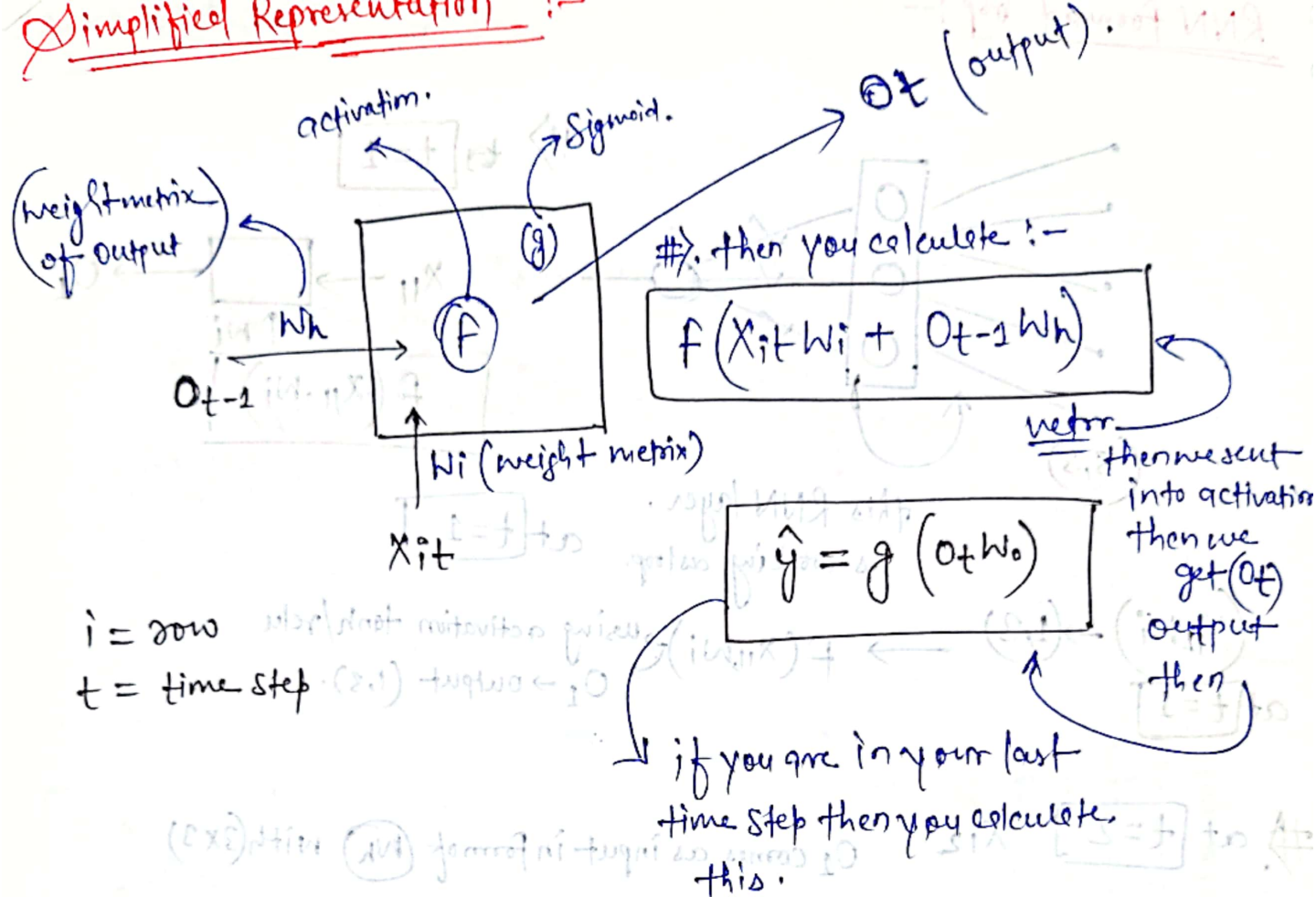
Scalar.

we put sigmoid on them.

Summary is on Review page.

A.T.O

Simplified Representation :-

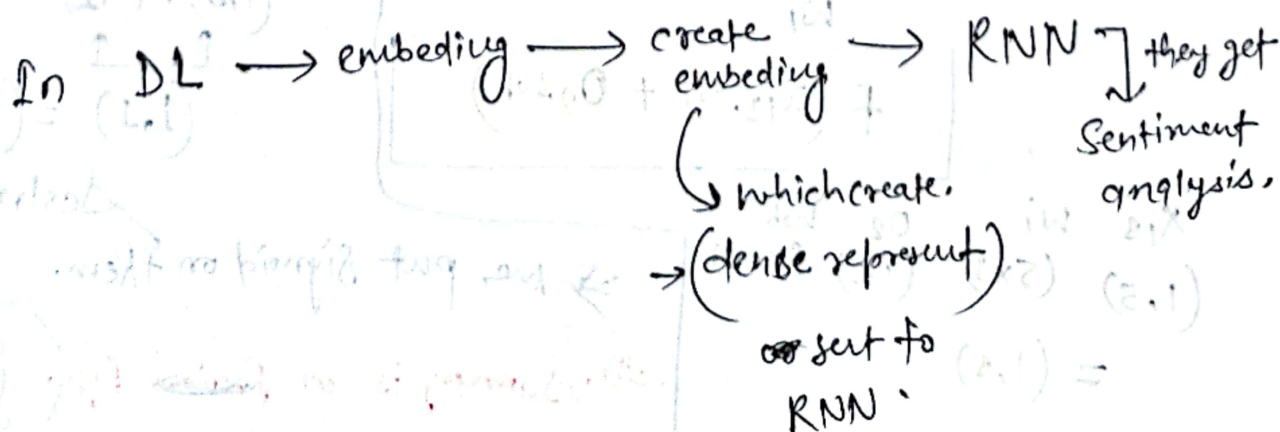


RNN Sentiment Analysis :-

⇒ Everything is available in code (How can we convert text into vectors and making models on them).

Embedding :-

In NLP (Natural language Processing) word embedding is a term used for the representation of word for text analysis typically in the form of a real valued vector and encodes the meaning of the words such that the words that are closer in vector space are expected to be similar in meaning.



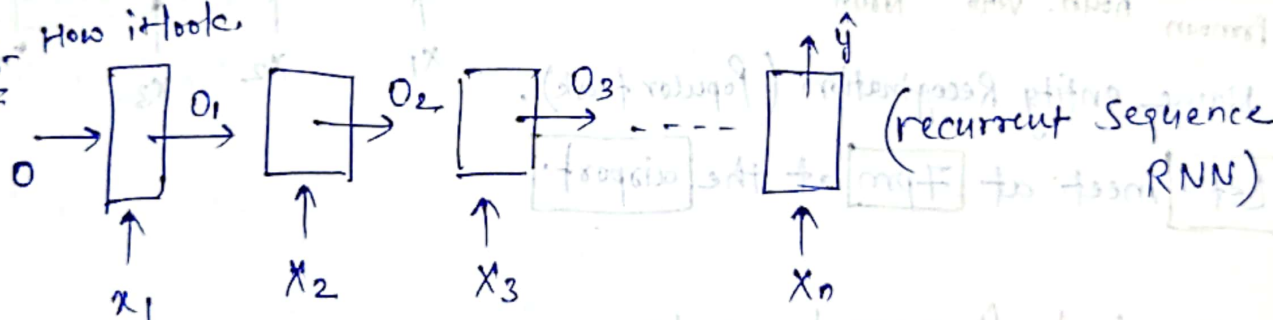
#7. Types of RNN

- ① Many to one
- ② One to many
- ③ Many to many
- ④ One to One.

① Many to One:-

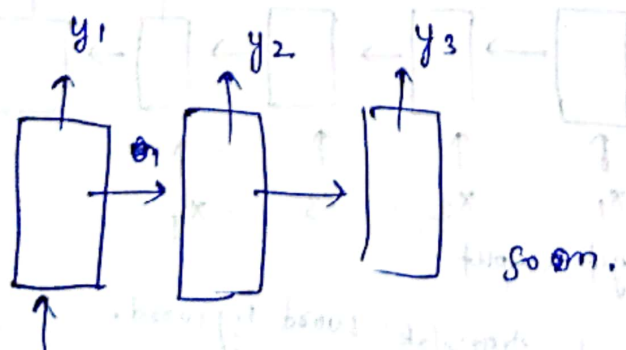
- It expects a sequence in input → ex:- sentence, character, time series
- It expects an output non-sequence → ex:- int/nom → scalar (1,0) depending on class.
- Ex:- Sentiment Analysis, Rating Prediction.
 - ↳ multiclassifⁿ (1-5).

Ex:- How it looks



② One to many:-

- in input normal non-sequential data get.
- in output we get sequential data.



→ where it use?

- image Captioning
- music generation.

Ex:-



results. Sequential.
A man is playing cricket.

#. Many to many

input $\rightarrow$ Sequential data.

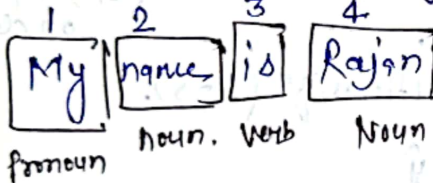
Output $\rightarrow$ Sequential data.

Two types :-

- ① Same length many to many
- ② Variable length many to many

① Same length many to many

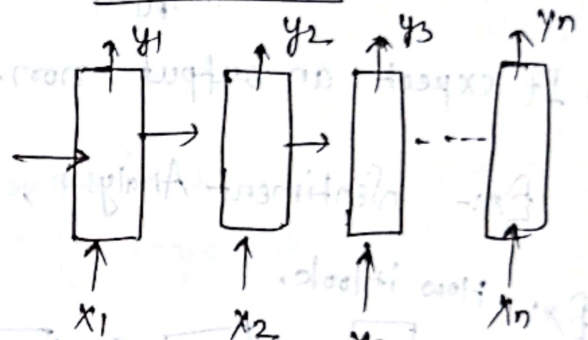
$\rightarrow$ we use in pos tagging $\rightarrow$ (NLP)



$\rightarrow$ Name entity Recognition (Popular task).

Let's meet at 7pm at the airport.

#. Flow diagram



② Variable length many to many :-

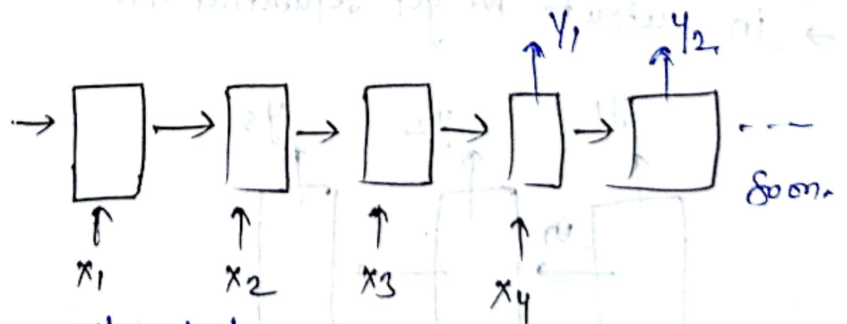
$\rightarrow$ Machine translation.

translate 1 language to 2nd language. (google translate).

Ex:-

Hi, my name is Rajon

मेरा नाम राजन है।



$\rightarrow$ why we put all input then next output.

When is in translation you can't do translate word by word.

this type of parts is called input $\rightarrow$ encoder.
output $\rightarrow$ decoder.

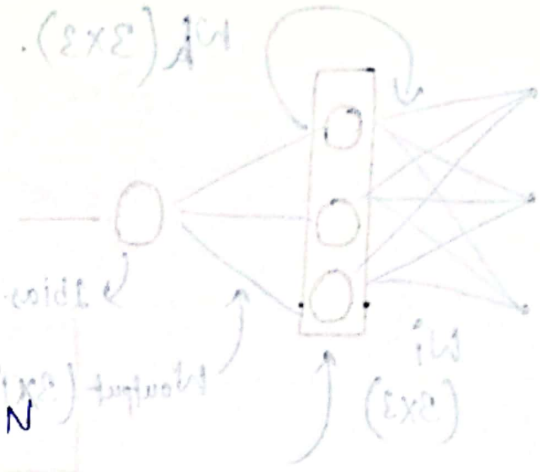
#). One to One :-

input $\rightarrow$ non sequential.
output $\rightarrow$ non sequential.

Ex:- use in Image classification.



this is simple NN

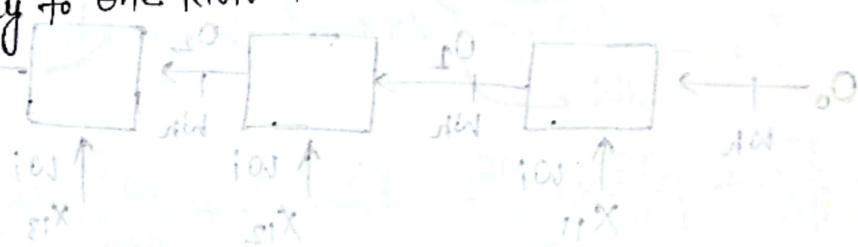


#). Backpropagation in RNN

$\Rightarrow$ we take an ex of many to one RNN to understand Backprop

Sentiment Analysis :-

text $\rightarrow$ 1/0



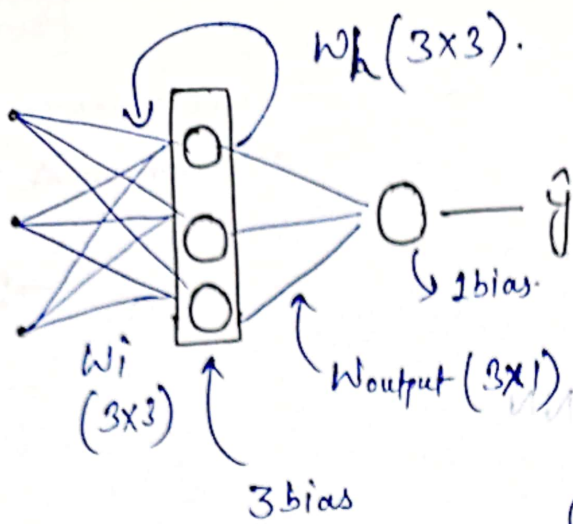
text reviews :-

Cat Cat rat
rat rat mat
Cat mat rat

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Vocab $\rightarrow$ Cat, mat, rat.
Cat $- [100]$
rat $- [010]$
mat $- [001]$

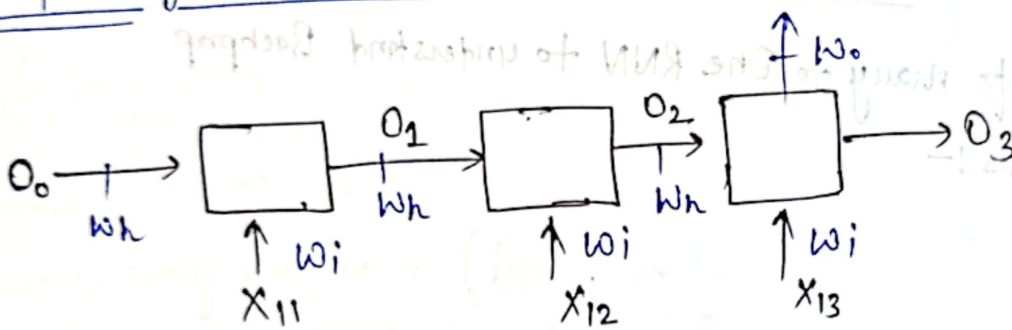
X_1 $[100], [100], [000]$ 1
 X_2 $[010], [010], [001]$ 1
 X_3 $[100], [001], [010]$ 0



Initial input nodes $\leftarrow$ input
Initial output nodes $\leftarrow$ output

$\Rightarrow$ you sent all vectors one by one, & you get output. O_1, O_2, O_3

#. Simple ways to understand:-



$$O_1 = f(X_{11} W_i + O_0 W_h)$$

$$O_2 = f(X_{12} W_i + O_1 W_h)$$

$$O_3 = f(X_{13} W_i + O_2 W_h)$$

$$\hat{y} = \sigma(O_3 \cdot W_o) \quad \text{activation (Sigmoid).}$$

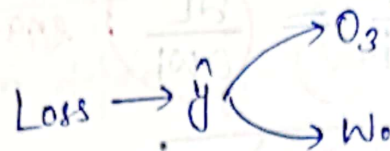
$$\text{Loss} = -y_i \log y_i - (1-y_i) \log (1-\hat{y}_i)$$

#. Once you calculating your loss then you should minimize value of W_i, W_h, W_o

$$\left\{ \begin{aligned} W_i &= W_i - \eta \frac{\partial L}{\partial W_i} \\ W_h &= W_h - \eta \frac{\partial L}{\partial W_h} \\ W_o &= W_o - \eta \frac{\partial L}{\partial W_o} \end{aligned} \right.$$

going from back to calculate

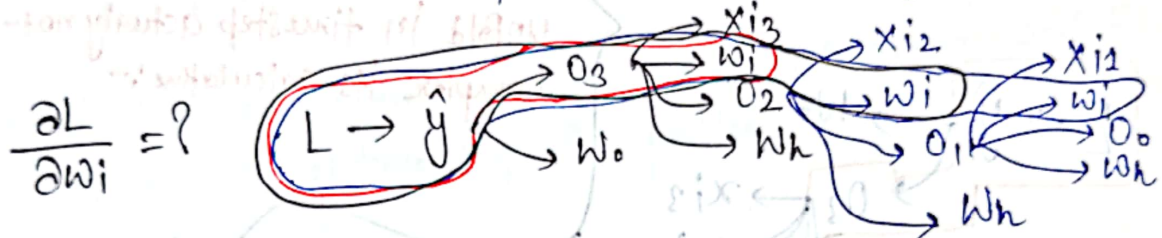
$$\frac{\partial L}{\partial w_0} = ?$$



$$\frac{\partial L}{\partial w_0} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_0}$$

$$\sigma(O_3 \cdot w_0)$$

from Loss we already calculate previous.



$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_3} \cdot \frac{\partial O_3}{\partial w_i} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial O_2} \times \frac{\partial O_2}{\partial w_i} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_3} \times \frac{\partial O_3}{\partial O_2} \times \frac{\partial O_2}{\partial O_1} \times \frac{\partial O_1}{\partial w_i}$$

$\Rightarrow$ Why 3 time bec'z our RNN folds 3 times. & also we have 3 words.

Summarize :-

$$\frac{\partial L}{\partial w_i} = \sum_{j=1}^3 \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_j} \cdot \frac{\partial O_j}{\partial w_i}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_1} \cdot \frac{\partial O_1}{\partial w_i} \Rightarrow \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_3} \cdot \frac{\partial O_3}{\partial O_2} \cdot \frac{\partial O_2}{\partial O_1} \cdot \frac{\partial O_1}{\partial w_i}$$

$$\Rightarrow \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_2} \cdot \frac{\partial O_2}{\partial w_i} \Rightarrow \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_3} \cdot \frac{\partial O_3}{\partial O_2} \cdot \frac{\partial O_2}{\partial w_i}$$

$$\cancel{\frac{\partial L}{\partial y}} \times \frac{\partial y}{\partial o_3} \times \frac{\partial o_3}{\partial w_i} = \frac{\partial L}{\partial w_i} \text{ exactly same.}$$

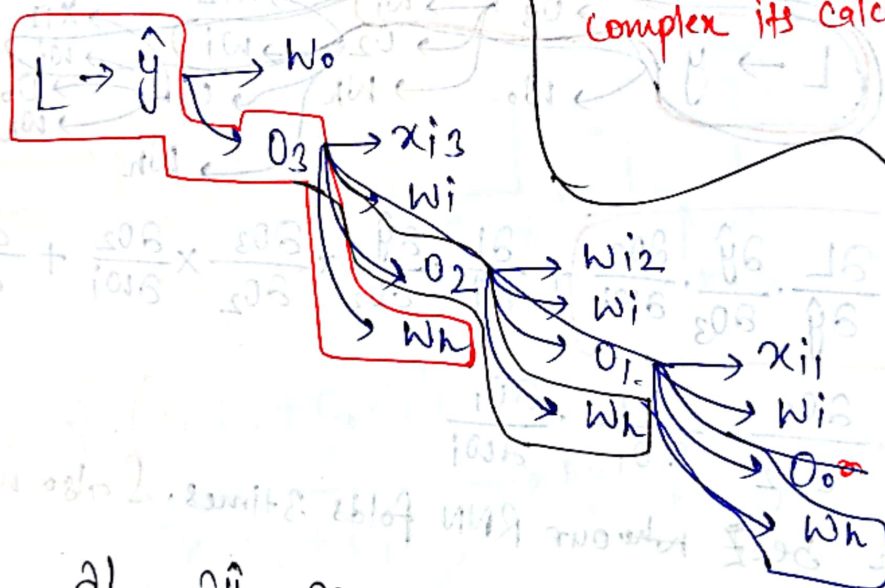
$$\frac{\partial L}{\partial w_i} = \sum_{j=1}^n \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_j} \cdot \frac{\partial o_j}{\partial w_i}$$

~~for n -step:~~

for n -times step.

$$\frac{\partial L}{\partial w_h} = ?$$

#7. We have to calculate backprop but it was too complex becz of unfold in timestep actually not complex its calculative:-



$$\frac{\partial L}{\partial w_h} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_3} \cdot \frac{\partial o_3}{\partial w_h} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_3} \cdot \frac{\partial o_3}{\partial o_2} \cdot \frac{\partial o_2}{\partial w_h} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial o_3}$$

$$\frac{\partial O_3}{\partial O_2} \cdot \frac{\partial O_2}{\partial O_1} \cdot \frac{\partial O_1}{\partial W_h}$$

Simplify :-

$$\frac{\partial L}{\partial \omega_n} = \sum_{j=1}^n \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial o_j} \cdot \frac{\partial o_j}{\partial \omega_n}$$

for n times. Step.

#>... Problems With RNN

- ① Problem of long term dependency. (Vanishing Gradient)
- ②. Problem of unstable training. / ~~(Vanishing Gradient)~~ (Exploding Gradient)
- ③. Stagnated training.

#. Example of long term dependency

Maharashtra is a beautiful place I went there last year but I want to enjoy properly but I don't becz I don't understand Marathi

⇒ this both word term Marathi depend on Maharashtra. and RNN fail in this type of time step. this problem is arise with vanishing Gradient problem.

#> Example with Stagnated training.

suppose you are training an RNN for sentiment analysis

Notice that after epochs 3 the loss barely changed. The model stagnates

Epochs	Training loss
1	1.20
2	0.95
3	0.89
4	0.83
5	0.83

Validation loss.

1.25
1.00
0.89
0.88
0.88

Problem 1 → Problem of long term dependency (Vanishing G.D)

RNN Structure

$$W_{input} = W_{input} - \eta \frac{\partial L}{\partial W_{input}}$$

$$W_{out} = W_{out} - \eta \frac{\partial L}{\partial W_{out}}$$

$$W_h = W_h - \eta \frac{\partial L}{\partial W_h}$$

Diagram illustrating the backpropagation through time (BPTT) for a recurrent neural network (RNN) structure, showing the calculation of gradients for weights W_{inp} and W_h across multiple time steps.

The diagram shows the gradient flow for $\frac{\partial L}{\partial W_{inp}}$ and $\frac{\partial L}{\partial W_h}$ across three time steps (1, 2, 3). The gradients are calculated using the chain rule, showing the contribution of each time step to the overall loss gradient.

Short term dependency is indicated for the first time step, and Long term dependency is indicated for the third time step.

Long term dependency

→ This will become very small to find $\frac{\partial L}{\partial W_{inp}}$ major contribution

is short term dependency.

(↑) time steps (↓) long term dependency

Soln to reduce this problem.

- ① Diffⁿ activation. → activation → use (relu/leaky relu)
- ② Better weight initialization.
- ③ Skip RNNs
- ④ LSTM

Problem 2.

(Viceversa of Problem 1)

if you are using $\tanh$ as activation, long term dependency become bigger to bigger. (W_{out} & W_h) become big and training stop.

Solⁿ

- ① Gradient clipping (you clip gradient value like no more increase value).
- ②. Controlled learning rate.
- ③. LSTM.

